

Mini Paper — 1.3 Exchanging Data

Time: ~35 minutes · Total: 30 marks · Answer all questions.

Mark your work against the mark scheme at the end; aim for ~1 minute per mark.

Questions

Q1. Explain what is meant by *hashing* and give **two** reasons why hashing is used to store passwords rather than storing the passwords directly. **[3]** (AO1)

Q2. A company sends confidential data over the internet.

(a) State the difference between *symmetric* and *asymmetric* encryption. **[2]** (AO1)

(b) Give **one** advantage of using asymmetric encryption when two parties have never communicated before. **[1]** (AO1)

(c) Explain why the company hashes its stored user passwords but encrypts the data it transmits, referring to the key difference between hashing and encryption. **[1]** (AO2)

Q3. A row of pixels in a monochrome image is shown below, where each letter is a pixel colour:

BBBBBBBWWWWBBBBBB

(a) Show how *run length encoding* (RLE) would encode this row. **[2]** (AO2)

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(b) State **one** type of data for which RLE would be a poor choice, and explain why. [1] (AO2)

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Q4. A flat-file table storing patient appointments is being normalised.

Appointment(AppointmentID, PatientID, PatientName, DoctorID, DoctorName, RoomNo)

Here DoctorName and RoomNo are determined by DoctorID.

(a) State the rule a table must satisfy to be in *first normal form (1NF)*. [1] (AO1)

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(b) The table is already in 2NF. Explain why it is **not** in *third normal form (3NF)*, and show the tables that result from normalising it to 3NF. [3] (AO2)

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Q5. A Member table has the fields MemberID, Surname, Town and JoinYear.

Write an SQL statement to display the MemberID and Surname of all members whose Town is 'Leeds' **and** who have a JoinYear before 2020, sorted by Surname in descending order. [4] (AO2)

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Q6. Describe the role of the **Transport** layer and the **Network/Internet** layer of the TCP/IP stack when data is sent. [3] (AO1)

Q7. Compare *packet switching* and *circuit switching*. [3] (AO2)

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Q8. An online banking website lets customers log in and transfer money between accounts.

[illegible]